

ZAP Release 0.2.1

Smart Contract Security Assessment

January 2026

Prepared for:

Meteora

Prepared by:

Offside Labs

Sirius Xie





Contents

1	About Offside Labs	2
2	Executive Summary	3
3	Detailed Summary	5
4	Disclaimer	6



1 About Offside Labs

Offside Labs is a leading security research team, composed of top talented hackers from both academia and industry.

We possess a wide range of expertise in modern software systems, including, but not limited to, *browsers*, *operating systems*, *IoT devices*, and *hypervisors*. We are also at the forefront of innovative areas like *cryptocurrencies* and *blockchain technologies*. Among our notable accomplishments are remote jailbreaks of devices such as the **iPhone** and **PlayStation 4**, and addressing critical vulnerabilities in the **Tron Network**.

Our team actively engages with and contributes to the security community. Having won and also co-organized *DEFCON CTF*, the most famous CTF competition in the Web2 era, we also triumphed in the **Paradigm CTF 2023** within the Web3 space. In addition, our efforts in responsibly disclosing numerous vulnerabilities to leading tech companies, such as *Apple*, *Google*, and *Microsoft*, have protected digital assets valued at over **\$300 million**.

In the transition towards Web3, Offside Labs has achieved remarkable success. We have earned over **\$9 million** in bug bounties, and **three** of our innovative techniques were recognized among the **top 10 blockchain hacking techniques of 2022** by the Web3 security community.



<https://offside.io/>



<https://github.com/offsidelabs>



https://twitter.com/offside_labs



2 Executive Summary

Introduction

Offside Labs completed a security audit of *Meteora ZAP* smart contracts, starting on *January 13th, 2026* and concluding on *January 13th, 2026*.

Project Overview

The ZAP program delivers one-click liquidity zaps. Zap Out enables users to exit LP positions or claim fees and immediately route a configured portion through whitelisted AMMs (e.g., DAMM V2, DLMM, Jupiter). Zap In lets users pre-record intended dual/single asset contributions in a ledger; the program then performs the necessary single sided swaps and adds or rebalances liquidity in DAMM V2 or DLMM. The goal is to collapse multi-step swap + LP workflows into a minimal set of on-chain calls, reducing operational overhead and slippage risk.

Updates Summary

Release 0.2.1 primarily adds support for DAMM v2 0.1.7. It also fixes the `zap_in` fee calculation and swap simulation for the newer base fee modes, and forwards the remaining accounts into the DAMM v2 `swap2` CPI to satisfy the rate limiter's single-swap validation.

Audit Scope

The assessment scope contains mainly the smart contracts of the zap program for the *Meteora ZAP* project.

The audit is based on the following specific branches and commit hashes of the codebase repositories:

- Meteora ZAP
 - Codebase: <https://github.com/MeteoraAg/zap-program>
 - PR-41:
 - Commit Hash: 020f3d23b114a662b946c384996a911a26811e60

We listed the files we have audited below:

- Meteora ZAP Release 0.2.1:
 - `programs/zap/src/instructions/ix_zap_in_damm_v2.rs`
 - `programs/zap/src/instructions/ix_zap_out.rs`
 - `programs/zap/src/instructions/zap_in_dlmm/ix_zap_in_dlmm_for_initialized_position.rs`
 - `programs/zap/src/instructions/zap_in_dlmm/ix_zap_in_dlmm_for_uninitialized_position.rs`
 - `programs/zap/src/lib.rs`
 - `programs/zap/src/utis/damm_v2_utils.rs`



Findings

The security audit identified no critical, high, medium, low, or informational issues in the specified PRs. The implementation adheres to security best practices for the assessed components.



3 Detailed Summary

Scope Confirmation

The audit covered:

1. Functionality: zap_in_damm_v2 now supports newer DAMM v2 fee modes and rate limiter validation, allowing the swap and liquidity add flow to.
2. PRs:
 - [PR-41](#)
 - Purpose: zap_in_damm_v2 now supports newer DAMM v2 fee modes and rate limiter validation, allowing the swap and liquidity add flow to execute on those pools.

Conclusion

After thorough review, the audited components:

- Follow secure design patterns for the intended functionality covered by the audit;
- Introduce no new security risks within the given threat model.



4 Disclaimer

This audit report is provided for informational purposes only and is not intended to be used as investment advice. While we strive to thoroughly review and analyze the smart contracts in question, we must clarify that our services do not encompass an exhaustive security examination. Our audit aims to identify potential security vulnerabilities to the best of our ability, but it does not serve as a guarantee that the smart contracts are completely free from security risks.

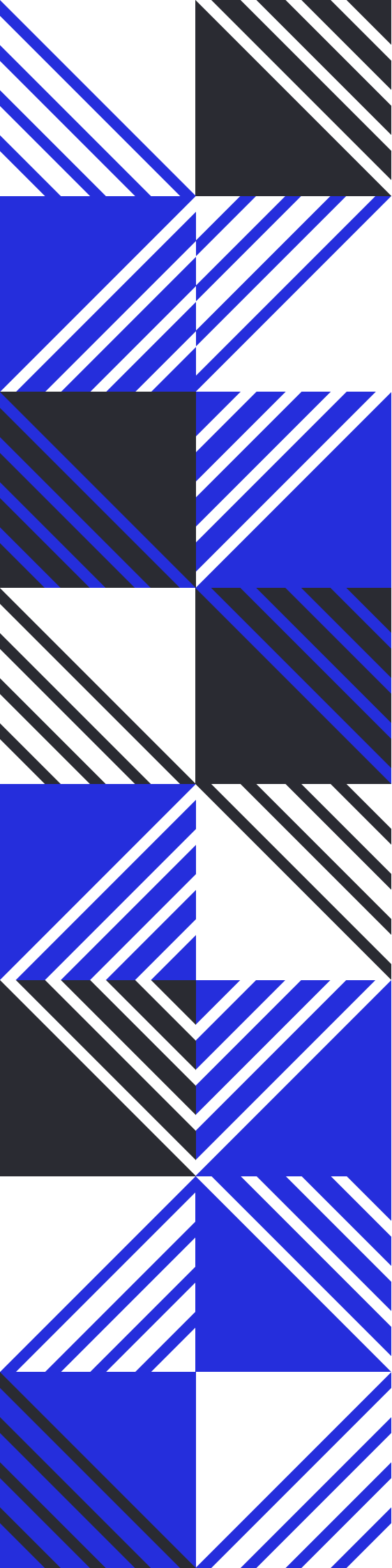
We expressly disclaim any liability for any losses or damages arising from the use of this report or from any security breaches that may occur in the future. We also recommend that our clients engage in multiple independent audits and establish a public bug bounty program as additional measures to bolster the security of their smart contracts.

It is important to note that the scope of our audit is limited to the areas outlined within our engagement and does not include every possible risk or vulnerability. Continuous security practices, including regular audits and monitoring, are essential for maintaining the security of smart contracts over time.

Please note: we are not liable for any security issues stemming from developer errors or misconfigurations at the time of contract deployment; we do not assume responsibility for any centralized governance risks within the project; we are not accountable for any impact on the project's security or availability due to significant damage to the underlying blockchain infrastructure.

By using this report, the client acknowledges the inherent limitations of the audit process and agrees that our firm shall not be held liable for any incidents that may occur subsequent to our engagement.

This report is considered null and void if the report (or any portion thereof) is altered in any manner.



 <https://offside.io/>

 <https://github.com/offsidelabs>

 https://twitter.com/offside_labs